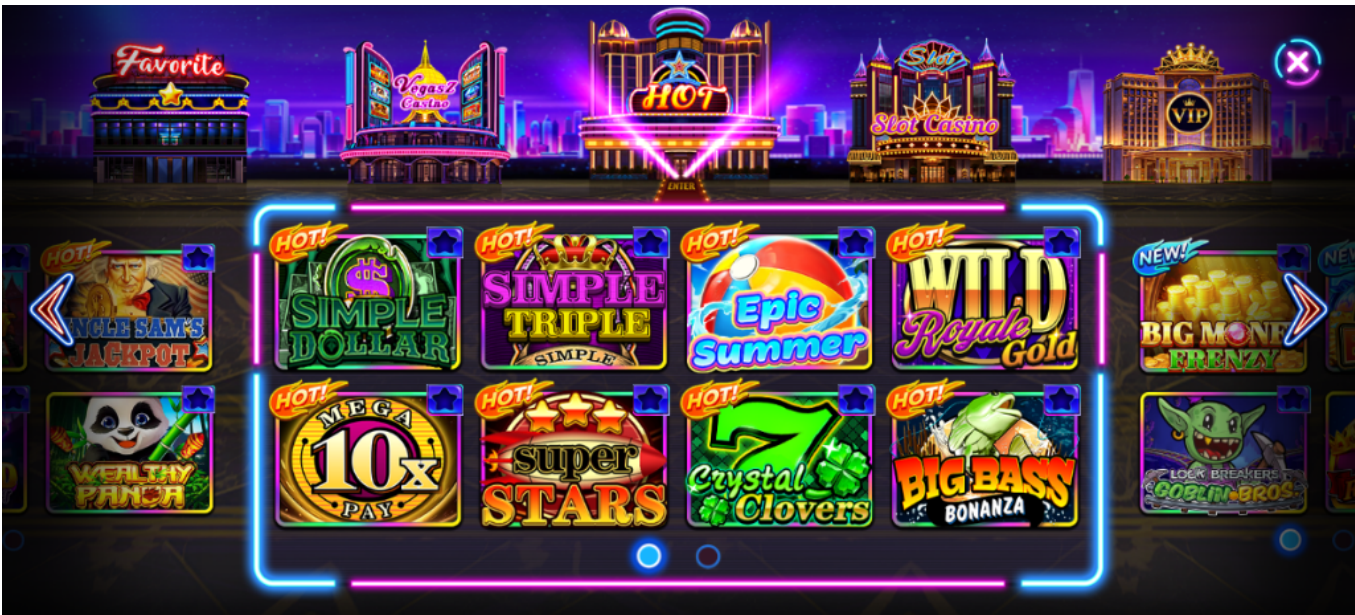


New Game Private Arcade

Administrator-controlled credits.
Android and iOS from the first release.

A Codex implementation handoff for original slots, keno and multiplayer fish tables. Updated to the owner's latest requirements on 17 September 2026.



Owner-supplied gameplay reference. This is not a new prototype or licensed release artwork.

CONTROL MODEL	PLAYER PLATFORMS	DELIVERY BACKLOG
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Manual admin ADD / REMOVE	Android + iOS	49 tickets / 7 sprints
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What changes now

Zero starting balances. No welcome credits, daily rewards, automatic refills or grant claims. Legitimate game winnings still settle automatically. The meaning of the slot's 30% target remains an explicit approval gate.

Delivery status: planning documents and tested reference code only. No app binary, deployed backend or running Codex task has been created. The 15-week staffed baseline is an estimate, not a promised completion date.

01 / CONFIRMED DIRECTION

The revised product baseline

This iteration replaces the earlier web-first and optional-refill assumptions. It preserves the private, no-actual-money context and the original three-game-family release.

REQUIREMENT	ITERATION 2
Main administrator	Explicit manual Add Credits and Remove Credits authority in the backend.
New accounts	Start at zero. Logging in or playing repeatedly never triggers a discretionary grant.
Automatic credits	No welcome, daily, promotional or scheduled credit allocations; no refill jobs.
Game winnings	Posted by the authoritative game settlement service. A win is not an admin grant.
Platforms	Installable Android and iOS player apps; a separate responsive admin web console.
Game catalog	One polished original slot, one keno game and one four-seat fish table.
Slot mathematics	No approved production profile. The earlier 96% illustration is not an owner-approved target.
Commercial boundary	No purchases, cash-outs, crypto or prizes with real-world value.

Proposed hierarchy

Main Admin → Sub-distributor → Agent → Player

The main/admin distributor is treated as the root. A separate main-distributor or owner tier is optional, not silently imposed.

Permission decision still needed

Can sub-distributors and agents move existing assigned credits inside their branch, or must every player addition/removal be performed by main admin? Until clarified, lower-tier credit mutations are disabled. No lower role inherits authority to create or remove supply.

02 / ADMINISTRATOR OPERATIONS

Add and remove safely

Main admin gets direct manual control without making the balance an editable database field. Every change is an attributable transaction, visible in both administrator audit history and the player's credit history.

STEP	REQUIRED BEHAVIOR
Select	Identify the exact account and branch. Display settled, reserved and available credits.
Prepare	Choose Add or Remove, enter a positive amount and a meaningful reason, and preview the result.
Confirm	Require recent privileged verification; bind the request to its target, amount and wallet version.
Commit	Recheck authority and availability, then atomically record balanced entries, audit data and revision.
Receipt	Return the durable transaction ID. Retries return that result instead of repeating the change.

Removal respects an accepted game action

Illustrative account: **100.00 settled, 20.00 reserved, 80.00 available**. Removing 90.00 is rejected. Removing 30.00 leaves 70.00 settled, 20.00 reserved and 50.00 available.

Main admin cannot remove the reserve, create a negative player balance or erase a settled win. An already accepted round may still award credits after the admin removes the currently available amount. Suspending new play does not cancel those awards.

No hidden corrective behavior

No direct Set Balance command; no automatic later collection of a rejected deduction; no history rewrite. Corrections use a new authorized, linked transaction. Stakes, payouts, admin changes and optional existing-credit transfers remain distinct event types.

Backend acceptance: duplicate-key and changed-body tests, stale-preview conflicts, concurrent remove/spin tests, exact integer arithmetic, audit/ledger atomicity and full reconciliation. The supplied pure helper tests do not replace these database tests.

03 / PAYOUT DECISION

Resolve the 30% meaning

“Around 30% payout with multiple attempts” can describe two materially different game designs. Neither interpretation has been selected in the production decision record.

INTERPRETATION	WHAT IT MEANS
A · Paying-spin frequency	Roughly 30 of 100 spins have some positive payout over a large sample. The size of those payouts, and therefore expected return, is a separate choice.
B · Credit return / RTP	Roughly 30 credits are returned per 100 credits staked in long-run expectation. This does not establish how many spins pay.
C · Another metric	Net-profitable rounds or sessions ending ahead would require their own precise definition and test model.

Two original illustrations, not product defaults

MODEL	PAYING SPINS	EXPECTED RETURN
70% at 0×; 22% at 2×; 6% at 4×; 2% at 14×	30%	96%
90% at 0×; 10% at 3×	10%	30%

The mathematics is direct: $0.22 \times 2 + 0.06 \times 4 + 0.02 \times 14 = 0.96$; $0.10 \times 3 = 0.30$. Both examples are explicitly named and require an explicit argument in the reference code. There is no implicit 96% runtime default.

Long-run target, not a session promise

A 30% paying-spin probability does not mean exactly three wins every ten spins. Do not force a quota or change an individual player's chances. At independent $p = 0.30$, ten nonpaying spins have probability 0.7^{10} , about 2.82%.

Before slot play can stake credits: approve the chosen metric, actual paytable, denominations, probability construction, version/hash and visible-grid evaluator. Keno and fish require separate rules; the slot target is not copied into them.

04 / IMPLEMENTATION ARCHITECTURE

One platform, two mobile apps

The proposed incremental route retains the earlier web-rendered game client and packages it with Capacitor for Android and iOS. Capacitor provides the native runtime; it does not supply the game engine, server or mathematics. [M01-M03]

LAYER	PROPOSED COMPONENT / RESPONSIBILITY
Player interface	Vue / TypeScript for lobby and navigation; PixiJS for original game scenes.
Native packaging	Capacitor Android + iOS projects. A shared web-rendered game inside an installable native shell.
Operator interface	Separate responsive Vue admin console. Not a privileged panel embedded in the player app.
Core services	NestJS identity, hierarchy, manual credit controls, rounds, reports and audit.
Multiplayer	Colyseus room/state service, using the same authoritative credit-settlement logic.
Durable data	PostgreSQL for ledger, identities, game results and recovery; immutable approved math versions.

The first sprint must prove the mobile choice

Test a representative slot, keno layout and dense moving fish scene on a real Android phone and iPhone. Record frame timing, memory behavior and background/resume recovery. A desktop browser mockup does not pass this gate.

If the shared renderer cannot meet the agreed device floor after a bounded optimization pass, present measured results and a scoped engine-change proposal before building the full game catalog.

Mobile correctness is part of the backend design

No new credit-staked actions while offline or backgrounded. Accepted actions still settle. Resume from the persisted result and wallet revision, not a queue of old touches. Test the same player switching between Android and iOS.

05 / NATIVE DELIVERY

Build and distribution decisions

Android and iOS are release deliverables now, not a “package it later” task. The owner's distribution route and build access determine the remaining external prerequisites.

ANDROID	iOS
Android Studio / SDK tooling and a reproducible signed build process. [M02,M13]	macOS with Xcode, locally or on a suitable runner, plus the owner-approved signing setup. [M02]
An APK is installable. An AAB is a publishing format processed into APKs by a distributor. [M05]	Produce and verify an Xcode archive; project generation alone is not a distributable signed app.
Private direct-install APK or a Play route, depending on the chosen release model. [M04]	TestFlight for beta; investigate an approved App Store route for ongoing use. [M06,M07]
Review current package-registration/developer-verification requirements before the actual release. [M12]	TestFlight builds have a 90-day test window; unlisted distribution requires approval and is not a private login boundary. [M06,M07]

Quality beyond a website wrapper

Original interactive games, mobile-safe layouts, platform back/close behavior, sound and motion settings, fast scene transitions and durable recovery are required. Apple evaluates original app-like functionality; using a native wrapper does not guarantee acceptance. [M08]

Truthful submission information

Review notes should describe the actual model: no credit purchases, no cash-out or prizes of value, and manual administrator credit assignment. Complete privacy and simulated-gambling/age-rating disclosures accurately for the intended markets. Do not assume “private” or “no money” removes review requirements. [M08–M10]

External prerequisites

Confirm private beta versus store release, target countries, minimum devices, Apple Developer account and Mac/macOS runner access. Keep account passwords, signing certificates and private keys out of chat and source control.

06 / SPRINT SEQUENCE

15-week staffed baseline

Planning assumption: two hands-on developers using Codex, part-time original-art/UI support and part-time QA/review. Re-estimate after native proof and the first complete game. Store review time is outside this engineering estimate. An owner-only workflow will likely take longer.

SPRINT	WEEKS	DELIVERABLE / GATE
S0	1-2	Native proof + foundation Android/iPhone rendering/lifecycle proof; reproducible setup; pending math gate.
S1	3-4	Manual admin controls Zero-start accounts; immutable ADD/REMOVE; no grant paths; concurrency tests.
S2	5-7	Both-platform slot alpha Approved math and exact visible payouts; complete admin-to-player journey on both platforms.
S3	8-9	Keno + reports Approved per-selection rules; recoverable draws; admin changes excluded from return.
S4	10-11	Four-seat fish table Independent model approval; single capture awards; failure/reconnect evidence.
S5	12-13	Polish + hardening Original assets; physical-device performance; security/load/restore reports.
S6	14-15	Beta + release preparation Owner-signed artifacts and approved distribution work; operational handover.

First end-to-end milestone: end of S2

Main admin creates a zero-credit player and adds credits. The player plays the original slot on Android and iOS. Main admin removes available credits. Reconnect and history show the same settled rounds and reconciled balance. No mandatory grant or transfer chain is inserted.

49 tickets define dependencies and acceptance evidence. Large tickets must be split before assignment. Extra titles, jackpots, automatic rewards and new operator tiers remain outside the baseline unless separately approved.

07 / ACCEPTANCE EVIDENCE

Correctness before expansion

GATE	REQUIRED PROOF
Credit integrity	Ledger rebuild matches balances. No duplicate changes, negative player balances, reserve theft or silent overwrites.
Manual-only policy	Create/login/date rollover/zero balance/device change never grants credits. No claim/refill jobs. Real game payouts remain valid.
Access control	Root-only ADD/REMOVE and branch-scoped lower roles. Forged IDs, exports, socket commands and retries cannot cross scope.
Math approval	No slot stakes while unresolved. Every slot grid matches its award. Keno/fish evaluated against their own approved models.
Native behavior	Real Android/iPhone lock, background, kill, resume, network switch, back/gesture and multi-device recovery tests.
Multiplayer	Only one award per target; stale room owner fenced; accepted actions survive failure without repeated charges.
Release readiness	Approved assets, compatible dependencies, signed-build provenance, privacy/rating information and restore/rollback drill.

Proposed performance targets, not measured results

Initial planning capacity: 50 simultaneous beta players and a recorded 100-user synthetic workload. Name the actual minimum Android and iPhone before promising support. Aim for smooth 60fps reference-device animation and a usable 30fps reduced-effects mode at the approved floor.

Run sustained fish scenes and repeated transitions for at least 20 minutes, collecting frame timing, memory and crashes. Record hardware, OS, build, network and asset versions. Browser screenshot tests complement, rather than replace, native testing.

Admin changes do not count as game return

Report manual additions/removals separately from stakes and game payouts. Show sample size and math version beside hit rate, net-winning-round frequency and payout/stake return. Do not label play credits as cash revenue.

08 / CODEX HANDOFF

Start with Sprint 0 only

The package contains revised repository instructions, decision gates and seven scoped prompts. Codex supports repository guidance through AGENTS.md; instructions still need to match the actual repository and verified commands. [M14]

Opening assignment

Paste into Codex after opening the extracted folder

Read AGENTS.md and the v2 change log, build spec, decisions, credit-control, mobile-delivery and sprint documents. Inspect existing code. Implement Sprint 0 only. Preserve manual main-admin ADD/REMOVE, zero initial credits and no automatic grants. Build Android/iOS project scaffolds and collect real-device proof. Keep slot math and lower-tier transfers disabled until owner approval. Return exact tests, files, screenshots and blockers; stop at the Sprint 0 review gate.

Commands that work in this handoff

node --test math-reference/game-math.test.mjs
policy-reference/product-policy.test.mjs

Run the two test paths together on one command line.
node math-reference/report.mjs

INCLUDED	PURPOSE
49-ticket backlog	JSON and readable Markdown; dependencies and acceptance criteria.
Machine-readable decisions	Confirmed no-grant/mobile requirements plus explicitly null slot targets/profile.
Manual credit policy reference	Pure posting validation; no live authentication, database or network writes.
Original math references	Explicit educational profiles and exact keno helpers; no implicit 96% default.
Seven prompts + source register	One sprint at a time; updated platform facts and source/asset review requirements.

Actual verification: 50 local reference tests passed; zero failures or skipped tests. This validates the supplied pure reference code and decision constraints only. It does not certify the platform, database, mobile performance, signing or store acceptance.

09 / OWNER DECISIONS

Three answers prevent rework

1. Which 30% target do you intend?

A: about 30 out of 100 spins pay something?

B: about 30 credits returned per 100 credits staked?

C: a different measure such as net-profitable rounds?

For A, payout sizes and expected return still need approval; 96% is not selected automatically.

2. Can lower tiers move existing credits?

Should sub-distributors and agents transfer only credits already assigned to them within their branch, or should all player adds/removes be main-admin-only? Until answered, the plan denies lower-tier credit mutations. Confirm separately if you need a main distributor distinct from the main admin.

3. How will Android/iOS apps be distributed?

Private installs/beta only, or Google Play/App Store release? Identify the group's countries and oldest phones, and whether a Mac/macOS runner and Apple Developer account are available. No keys or passwords are required here.

Work that can begin before all answers

Workspace scaffolding, zero-credit accounts, scoped permissions, administrator adjustment screens, generic math reports, original interface prototypes and Android/iOS performance proof. Slot staking waits for actual math approval; only that feature is blocked by the payout ambiguity.

Later feature decisions

Keno draw count/paytables, fish capture versus damage rules, final branding and concurrency/hosting expectations remain explicit feature gates. They do not justify inventing unseen JUWA algorithms. No additional broad source-code search is needed before S0 begins.

This iteration assumes the previously stated no-actual-money model remains unchanged. It is a technical product plan, not an app-store approval or jurisdiction-specific legal determination.

Primary research references

Official documentation consulted for iteration 2 on 17 September 2026. Platform rules and toolchains must be checked again when pinning versions and preparing release. No SDK/device execution or external approval was performed during this research.

M01 • Capacitor cross-platform runtime

Shared web application packaged for Android/iOS; proposed addition to the v1 stack.

<https://capacitorjs.com/>

M02 • Capacitor environment setup

macOS/Xcode for iOS; Android Studio/SDK for Android; pin exact compatible tooling in S0.

<https://capacitorjs.com/docs/getting-started/environment-setup>

M03 • Capacitor source and MIT license

Root license verified for research; preserve notices and recheck at selected revision.

<https://raw.githubusercontent.com/ionic-team/capacitor/main/LICENSE>

M04 • Android alternative distribution

Direct APK and marketplace options; route needs owner approval.

<https://developer.android.com/distribute/marketing-tools/alternative-distribution>

M05 • Android bundle FAQ and signing

APK installable versus AAB publishing format; signing requirements referenced separately.

<https://developer.android.com/guide/app-bundle/faq>

M06 • TestFlight overview

90-day build test duration; review and tester distribution.

<https://developer.apple.com/help/app-store-connect/test-a-beta-version/testflight-overview/>

M07 • Unlisted app distribution

Approved limited-audience distribution candidate; link discoverability is not login authorization.

<https://developer.apple.com/support/unlisted-app-distribution/>

Primary research references

Official documentation consulted for iteration 2 on 17 September 2026. Platform rules and toolchains must be checked again when pinning versions and preparing release. No SDK/device execution or external approval was performed during this research.

M08 • Apple App Review Guidelines

Original app-like functionality, truthful metadata/review access and policy review; no acceptance guarantee.

<https://developer.apple.com/app-store/review/guidelines/>

M09 • Apple age ratings

Disclose simulated gambling accurately; country and rating review before release.

<https://developer.apple.com/help/app-store-connect/reference/app-information/age-ratings-values-and-definitions>

M10 • Capacitor privacy manifest

Review used APIs/plugins and required reasons; not a claim all plugins need the same manifest.

<https://capacitorjs.com/docs/ios/privacy-manifest>

M11 • Capacitor app lifecycle API

Lifecycle events for background/resume; durable game recovery remains application work.

<https://capacitorjs.com/docs/apis/app>

M12 • Android developer verification

Current identity/package registration guidance for out-of-Play distribution; review rollout and applicable region.

<https://developer.android.com/developer-verification/guides/android-developer-console>

M13 • Android signing

Signed release APKs/AABs and protection of release keys.

<https://developer.android.com/studio/publish/app-signing>

M14 • Codex repository instructions

AGENTS.md workflow; official address currently redirects to ChatGPT Learn.

<https://developers.openai.com/codex/guides/agents-md/>

Carried-forward implementation research

The full kit retains the v1 register for Pixijs, Colyseus, Vue Pure Admin, NestJS, Engine math SDK, keno references, testing and asset sources. Those entries are research leads, not security audits. Recheck the chosen revision, license and every bundled asset before reuse.